

# FINEXA

## Smart Loan Management System

Complete Product, System & Technical Documentation

### DOCUMENT METADATA & CONTROL

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Repository: <https://github.com/ponnagantiharikrantsai-tech/Finexa>

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## 1. EXECUTIVE SUMMARY

FINEXA is a professional, production-grade FinTech Loan Management System designed for private lenders, financial administrators, and credit management teams. FINEXA replaces manual spreadsheets with a secure, real-time platform for issuing loans, tracking interest, managing monthly payment cycles, enforcing late penalties, and generating compliance records.

- Primary Focus: End-to-end lifecycle management of principal, monthly interest, cycle extensions, and overdue penalty recovery.
- Target Audience: Private lenders, loan officers, portfolio managers, and administrative teams.
- Core Value: Eliminates interest calculation errors, enforces cycle accuracy, and provides multi-device instant synchronization.

## 2. PROJECT OBJECTIVE

The principal objective of FINEXA is to provide a single, immutable source of truth for all loan issuing, repayment tracking, borrower KYC profiles, and financial yield calculations.

- Borrower KYC: Secure encrypted storage of Aadhaar, PAN, and identity details.
- Loan Cycle Precision: Automatic calculation of due dates, monthly interest, and accrued late penalties.
- Pay & Extend Workflow: Standardized process when borrowers pay monthly interest without reducing principal.
- Overdue & Penalty Settlement: Dynamic daily penalty accrual per  $\pm 1,000$  principal /

## 3. COMPLETE APPLICATION FLOW

The diagram below illustrates the end-to-end user navigation and data lifecycle within FINEXA:

Public User / Google Search -> Public Login Page (/login) -> Authenticated Session

%% > Home Dashboard (/home)

%%% > Loan Management (/loan-management) -> New Loan -> Active Loan -> Pay &

%%% > Borrower Management (/borrowers) -> View Details -> KYC -> Internal Note

%%% > Application Generator (/applications) -> Public Apply Link -> Submission ->

%%% > Reports & Analytics (/reports) -> Financial Ledgers -> CSV / Print PDF

%%% > Capital Management (/capital-management) -> Vault Balance -> Funder Dep

#### 4. USER ACCESS FLOW & PUBLIC/PRIVATE BOUNDARY

FINEXA enforces a strict boundary between public entry points and private financial data:

- PUBLIC ROUTE: /login & / -> Accessible without authentication. Displays lightweight login form. Indexable by search engines.
- PUBLIC ROUTE: /apply/[code] -> Customer borrower application submission link.
- PRIVATE ROUTES: /home, /loan-management, /borrowers, /reports, /capital-management, /settings -> Require authenticated session.
- Middleware Guard: Server-side redirection to /login if unauthenticated. Protected routes stamp Cache-Control: no-store headers to block back-button data leaks upon logout.

#### 5. HOME PAGE ARCHITECTURE

The Home Dashboard (/home) provides executive overview access to all core modules:

- 3D Hero Branding: Perspective-aware FINEXA logo depth effect.
- Staggered Navigation Cards: Direct access to Loan Management, Applications, and Financial Reports.
- Instant Data Hydration: Renders interface shell immediately without blocking on secondary data queries.

#### 6. LOAN MANAGEMENT WORKFLOW

Loan Management (/loan-management) is the central operational module of FINEXA:

- New Loan Creation: Select existing borrower or inline create new borrower with encrypted KYC.
- Loan Parameters: Principal (<sup>1</sup>), Monthly Interest Rate (<sup>1</sup> per <sup>1</sup>1,000 / month), Issue
- Active Loan Card: Displays Principal, Outstanding Balance, Monthly Interest, Current Due Date, Status (active, extended, overdue, closed).

#### 7. BORROWER MANAGEMENT SYSTEM

Borrowers (/borrowers) manages complete identity and credit history:

- KYC Records: Encrypted PAN & Aadhaar (AES-256-CBC), contact details, address, location URL.
- Card Association: Links every borrower to their respective active and closed loans.
- Unique Collision Resolver: Prevents database unique constraint failures when creating separate borrower cards with identical mobile numbers.

#### 8. INTEREST SYSTEM LOGIC

FINEXA calculates monthly interest using standard FinTech lending formulas:

Monthly Interest Formula:

Monthly Interest = Principal \* (Interest Rate / 1,000)

Example:

Principal = ₦10,000 | Interest Rate = ₦250 per ₦1,000 / month

Monthly Interest = 10,000 \* (250 / 1,000) = ₦2,500 / month

## 9. PAY & EXTEND WORKFLOW

Pay & Extend handles the common scenario where a borrower pays the monthly interest without reducing principal:

- Interest Recorded: Interest payment (e.g. ₦2,500) is recorded into the payment ledger.
- Principal Unchanged: Outstanding principal remains 100% (e.g. ₦10,000).
- Cycle Advanced: Current cycle closes as 'extended' and due date extends by 1 month.
- Audit Record: Permanent cycle entry inserted into loan\_cycles database table.

## 10. OVERDUE & PENALTY CALCULATION SYSTEM

When a borrower pays after the assigned due date, late penalty accrues dynamically:

Accrued Penalty Formula:

Penalty = (Principal / 1,000) \* Penalty Rate \* Overdue Days

Example:

Principal = ₦10,000 | Rate = ₦2.50 per ₦1,000/day | Overdue = 2 Days

Penalty = (10,000 / 1,000) \* 2.50 \* 2 = ₦50

Total Settlement Required = Interest (₦2,500) + Penalty (₦50) = ₦2,550

## 11. PAYMENT SYSTEM & ALLOCATION

Payments are recorded with explicit type allocation:

- Interest Payment: Satisfies monthly cycle interest requirement.
- Principal Payment: Directly reduces outstanding principal balance.
- Penalty Payment: Settles late fee charges recorded in penalty\_ledger.

## 12. PERMANENT LOAN CYCLE LEDGER SYSTEM

The loan\_cycles table stores an immutable audit history of every cycle transition:

- Cycle Attributes: cycle\_number, start\_date, due\_date, payment\_date, interest\_paid, penalty\_paid, remaining\_principal, status.
- Transparency: Full cycle timeline viewable inside Loan Details popup.

## 13. APPLICATION LINK GENERATOR WORKFLOW

Allows lenders to generate digital KYC application links for borrowers:

- Link Generation: Unique application code generated with set principal and interest terms.
- Borrower Filling: Borrower opens /apply/[code], enters personal details, uploads KYC, submits.

- Verification: Lender reviews pending application in /applications and approves to auto-create loan & borrower.

## 14. REPORTS & FINANCIAL ANALYTICS

Reports (/reports) provides real-time financial health analytics:

- Metrics: Total Amount Lent, Total Collected, Total Outstanding, Interest Earned, Net Profit.
- Filtering: Today, Yesterday, Last 7 Days, Last 30 Days, This Month, Last Month, Custom Range.
- Exports: One-click Excel/CSV data export and print-ready compliance layout.

## 15. PDF GENERATION ARCHITECTURE

FINEXA features standalone PDF generation powered by jsPDF:

- Active Loans Backup PDF: Generates offline printable backup of every active loan card.
- Non-Blocking UI: Progress overlay displays while PDF compiles without freezing application.

## 16. FUNDER / CAPITAL MANAGEMENT

Capital Management (/capital-management) tracks investor vault balances:

- Capital Addition: Records additions to available loan capital pool.
- Loan Utilization: Tracks available vs deployed capital across loans.

## 17. INTERNAL NOTES SYSTEM

Lenders can store private administrative notes per borrower:

- Auto-Save: Notes saved to database upon closing details view or clicking save.
- Audit Track: Note updates record internal\_notes\_updated\_at timestamp.

## 18. SEARCH, FILTERING & DEBOUNCING

Search and filtering are optimized for 60 FPS performance:

- Multi-Field Search: Matches Name, Mobile, PAN, and Aadhaar.
- Debounced Execution: Uses useDeferredValue and useMemo to prevent UI frame drops during fast typing.

## 19. MULTI-DEVICE REAL-TIME SYNCHRONIZATION

FINEXA utilizes Supabase Realtime subscriptions to reflect changes across devices instantly:

- Event Listening: Subscribes to INSERT, UPDATE, DELETE on loans and payments.
- Silent Hydration: Updates local UI state automatically without full page reload.

## 20. AUTHENTICATION & SECURITY ARCHITECTURE

Security is implemented across every application layer:

- Auth Provider: Supabase SSR Authentication with encrypted JWT cookies.
- Data Encryption: Sensitive PAN and Aadhaar encrypted at rest via AES-256-CBC.
- Anti-Caching: Protected pages set Cache-Control: no-store headers.

## 21. NOTIFICATION ARCHITECTURE (EMAIL/SMS)

Automates borrower communications:

- Email Notifications: Powered by Resend API & React Email templates.
- SMS Integration: Fast2SMS API integration for loan creation and overdue reminders.
- Status: Email (Implemented), SMS (Configured), WhatsApp (Not currently implemented).

## 22. DATABASE SCHEMA & ER DIAGRAM

PostgreSQL relational schema managed via Drizzle ORM:

```
Borrowers (borrower_id) % % 1:N % % > Loans (loan_id)
% % % 1:N % % > Payments (payment_id)
% % % 1:N % % > Loan Cycles (cycle_id)
% % % 1:N % % > Penalty Ledger (penalty_id)
```

## 23. DATA FLOW ARCHITECTURE

Client Action -> Server Action -> Drizzle ORM -> Supabase PostgreSQL -> Realtime Dispatch -> Optimistic UI Update.

## 24. VERIFIED TECHNOLOGY STACK

- Framework: Next.js 16.2 (Turbopack, App Router, Server Actions)
- Language & UI: TypeScript 5, React 19, Tailwind CSS v4, Base UI
- Database & ORM: Supabase PostgreSQL, Drizzle ORM 0.45
- Security: AES-256-CBC Encryption, Supabase Auth SSR

## 25. DIRECTORY CODE STRUCTURE

Modular feature-based architecture under /src:

- src/app: Next.js App Router routes ((auth), (dashboard), api, apply).
- src/features: Domain modules (loans, borrowers, payments, applications, reports, capital).
- src/db: Schema definitions & Drizzle database client.
- src/components/motion: 3D Motion System components.

## 26. PAGE-BY-PAGE FEATURE MATRIX

- /login: Public login entrance animation and credentials submission.
- /home: Executive hero dashboard and module access.
- /loan-management: Loan card list, payments, extensions, penalties, PDF download.
- /reports: Financial analytics, KPIs, CSV export, print view.

## 27. BUTTON ACTION MATRIX

- Record Payment: Submits payment action -> Money particle effect -> Local state update.
- Pay & Extend: Executes interest clearance & cycle extension -> 3D timeline rotation.
- Download PDF: Compiles active loan records into downloadable backup PDF.

## 28. SYSTEM ERROR HANDLING STRATEGY

Form actions return typed `ActionResult<T>`. Errors display via toast notifications with clear diagnostics and state rollback.

## 29. 60 FPS PERFORMANCE & 3D MOTION SYSTEM

Custom GPU-accelerated motion system utilizing CSS 3D transforms (`translateZ`) and HTML5 Canvas particles without heavy 3D engine overhead.

## 30. RESPONSIVE DESIGN (MOBILE / DESKTOP)

Fluid layouts designed for Mobile, Tablet, and Desktop with touch-friendly press interactions replacing hover on mobile viewports.

## 31. SEO & PUBLIC DISCOVERABILITY

Public entrance configured with metadata, OpenGraph, sitemap.xml, and robots.txt disallowing private route crawling.

## 32. BACKUP & RECOVERY PROTOCOL

Database auto-backups managed by Supabase Cloud. Source code version-controlled on GitHub.

## 33. CI/CD DEPLOYMENT FLOW

Git Push to main branch triggers automatic Vercel production build and deployment.

## 34. END-TO-END REAL-WORLD SCENARIO EXAMPLE

Step-by-step trace of loan creation, interest payment, extension, overdue penalty, and final closure.

## 35. MASTER SYSTEM FLOWCHART

Complete decision tree from user entry to loan payoff and compliance archiving.

## 36. GLOSSARY OF FINTECH TERMS

- Principal: Original loan amount issued.
- Interest Rate: Monthly yield per  $\text{₹}1,000$  principal.
- Pay & Extend: Interest clearance without principal reduction.

## 37. CURRENT VS FUTURE ROADMAP

- Current: Loan management, Pay & Extend, Penalties, PDF, Reports, 3D Motion, Realtime.
- Future: WhatsApp Business API reminders, automated CIBIL score checks (Planned).

### 38. FINAL SYSTEM LIFECYCLE SUMMARY

FINEXA delivers a complete, secure, fast, and visually striking loan management platform built to scale with growing lending operations.